4.14

The lifetime of a lightbulb can be modeled with an exponential random variable with an expected lifetime of 1000 days.

a

Find the probability that the lightbulb will function for more than 2000 days.

✓ **Answer**

$P\left(\mathrm{Exp}\left(\frac{1}{1000}\right) > 2000\right) = 13.53352832366127\%$

b

Find the probability that the lightbulb will function for more than 2000 days given that it is still functional after 500 days.

✓ **Answer**

Since the exponential distribution is memoryless, this is equivalent to

$P\left(\mathrm{Exp}\left(\frac{1}{1000}\right) > 1500\right) = 22.31301601484298\%$

4.49

Suppose that you own a store that sells a particular stove for $\$1000$. You purchase the stoves from the distributor for $\$800$ each. You believe that this stove has a lifetime which can be faithfully modeled as an exponential random variable with a parameter of $\lambda = 1/10$, where the units of time are years.

You would like to offer the following extended warranty on this stove: if the stove breaks within r years, you will replace the stove completely (at a cost of $\$800$ to you). If the stove lasts longer than r years, the extended warranty pays nothing.

Let $\$C$ be the cost you will charge the consumer for this extended warranty. For what pairs of numbers (C, r) will the expected profit you get from this warranty be zero? What do you think are reasonable choices for C and r? Why?

✓ **Answer**

$X \sim \mathrm{Exp}(\frac{1}{10})$

$E_C(r) = 800(P(X < r))$

Would give us the following expected costs:

1 years: 76.1300655712324
2 years: 145.01539753761455
3 years: 207.3454234546257
4 years: 263.74396317148853
5 years: 314.77547222989324
6 years: 360.95069112477887
7 years: 402.73175696687247
8 years: 440.53682870622276
9 years: 474.74427220752074
10 years: 505.6964470628461

I believe a reasonable (C, r) would be $(210, 3)$

5.4

In parts a-d below, either use the information given to determine the distribution of the random variable, or show that the information given is not sufficient by describing at least two different random variables that satisfy the given condition.

a

X is a random variable such that $M_X(t) = e^{6t^2}$ when $|t| < 2$.

✓ **Answer**

This appears to be a Normal distribution.

$X \sim N(0, 12)$

b

Y is a random variable such that $M_Y(t) = \frac{2}{2-t}$ for $t < 0.5$.

✓ **Answer**

This appears to be an exponential distribution.

$Y \sim \text{Exp}\,(\lambda = 2)$ (or $\beta = 0.5$ depending on your choice of parameter)

c

Z is a random variable such that $M_Z(t) = \infty$ for $t \geq 5$.

✓ **Answer**

Not enough information.

d

W is a random variable such that $M_W(2) = 2$.

✓ **Answer**

This just says that the $E[X^2] = 2$, which gives us no information about the distribution.

5.8

Let $X \sim \text{Unif}[-1, 2]$. Find the probability density function of the random variable $Y = X^2$.

✓ **Answer**

$f_x(x) = \begin{cases} \frac{1}{3} & -1 \leq x \leq 2 \\ 0 & \text{else} \end{cases}$

$X = \pm\sqrt{Y}$
$\frac{dX}{dY} = \frac{1}{2\sqrt{Y}}$

$f_y(y) = \begin{cases} \frac{1}{3\sqrt{y}} & 0 < y \leq 1 \\ \frac{1}{6\sqrt{y}} & 1 < y \leq 4 \\ 0 & \text{else} \end{cases}$

5.11

The random variable X has the following probability density function:

$$f_X(x) = \begin{cases} xe^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

a

Find the moment generative function of X.

✓ **Answer**

$M_X(t) = E[e^{xt}]$
$= \int\limits_0^\infty xe^{-x(1-t)}\, dx$
$= \frac{\Gamma(2)}{(1-t)^2} = (1-t)^{-2}$

b

Using the moment generating function of X, find $E[X^n]$ for all positive integer n. Your final answer should be an expression that depends only on n.

✓ Answer

$m(n) = (n+1)!$

5.18

Let $X \sim \text{Geom}(p)$.

a

Compute the moment generating function $M_X(t)$ of X. Be careful about the possibility that $M_X(t)$ might be infinite.

✓ Answer

$M_X(t) = E[e^{xt}]$

$\sum_{x=1}^{\infty} e^{xt} p(1-p)^{x-1}$

$= pe^t \sum_{x=0}^{\infty} (e^t(1-p))^x$

$= \frac{pe^t}{1-(1-p)e^t}$

Where $t < -\ln(1-p)$

b

Use the moment generating function to compute the mean and the variance of X.

✓ Answer

$M_X'(t) = \frac{pe^t}{(1-(1-p)e^t)^2}$

$\Big|_0 = \frac{1}{p}$

$M_X''(t) = \frac{pe^t - p^2e^t + p}{(1-e^t(1-p))^3}$

$\Bigg|_0 = \frac{2-p}{p}$

$\text{Var}(X) = \frac{1-p}{p}$

5.35

The floor function $\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$ gives the largest integer less than or equal to x. Let $X \sim \text{Exp}(\lambda)$. Find the probability mass function of $\lfloor x \rfloor$.

✓ **Answer**

$p_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{else} \end{cases}$

$Y = \lfloor X \rfloor$

$p_Y(y) = \int_y^{y+1} \lambda e^{-\lambda x}\, dx = e^{-\lambda y}(1 - e^{-\lambda})$

$Y \sim \text{Geom}(1 - e^{-\lambda})$

6.3

For each lecture the professor chooses between white, yellow, and purple chalk, independently of previous choices. Each day she chooses white chalk with probability 0.5, yellow chalk with probability 0.4, and purple chalk with probability 0.1.

a

What is the probability that over the next 10 days she will choose white chalk 5 times, yellow chalk 4 times, and purple chalk 1 time?

✓ **Answer**

$\frac{10!}{5!4!1!}(0.5)^5(0.4)^4(0.1) \approx 0.10080000000000003$

b

What is the probability that over the next 10 days she will choose white chalk exactly 9 times?

✓ **Answer**

$X \sim \text{Binom}(10, 0.5)$

$P(X = 9) = 10(0.5)^{10} \approx 0.009765625$